Draft tables: watermark stress test

Sources read by this notebook:

Table in the draft	File
Table 4 (survival matrix)	`aseal/audioseal_results.csv`, `wavmark/wavmark_results.csv`, `audiowmark/audiowmark_results.csv`, `perth/claude_run_2026-11-10/perth_mono_results.csv`, `craw/craw_results_30s.csv`, `synthid_results.csv`, newest `aware/runs/*/results.csv`
Table 5 (Wilson intervals)	computed
Table 6 (ViSQOL)	`aseal/visqol_results.csv`
Appendix B (AWARE by song)	newest `aware/runs/*/results.csv` and `embedding.json`

In [ ]:
```
import json, math
from pathlib import Path
import pandas as pd

ROOT = Path.cwd()
if not (ROOT / "attacks.py").exists():          # app opened elsewh
    ROOT = Path.home() / "Downloads" / "watermark-stress-test-main"
print("Repository folder:", ROOT)
aware_run = sorted((ROOT / "aware" / "runs").glob("2*"))[-1]   # ne
print("AWARE run used:", aware_run.name)

def yes(series):
    return series.astype(str).str.lower().eq("true")
```

```
Repository folder: /Users/antoniataubert/Downloads/watermark-stress-
test-main
AWARE run used: 20261110_183744_325036
```

Load every tool's results

In [2]:
```
aseal = pd.read_csv(ROOT / "aseal/audioseal_results.csv")
wavmark = pd.read_csv(ROOT / "wavmark/wavmark_results.csv")
awm = pd.read_csv(ROOT / "audiowmark/audiowmark_results.csv")
perth = pd.read_csv(ROOT / "perth/claude_run_2026-11-10/perth_mono_
craw = pd.read_csv(ROOT / "craw/craw_results_30s.csv")
synthid = pd.read_csv(ROOT / "synthid_results.csv")
```

```
aware = pd.read_csv(aware_run / "results.csv")
print({name: len(df) for name, df in
       dict(audioseal=aseal, wavmark=wavmark, audiowmark=awm, perth
```

```
{'audioseal': 104, 'wavmark': 88, 'audiowmark': 88, 'perth': 112, 'c
raw': 48, 'synthid': 37, 'aware': 104}
```

Table 4. Survival matrix

Each cell is *songs where it worked / songs tested*. "Detected" uses the tool's own default threshold; "exact" means every payload bit was recovered. On the control row (unmarked originals), anything above 0 is a false alarm. Rows with status `error` (stem split without Demucs) are left out.

In [3]:

```
EDITS = ["none", "control_unmarked", "mp3128", "aac128", "gain6db",
         "polarity", "stemsplit", "timestretch5", "speedup10", "slo

def count(df, flag_col, status_col="status"):
    ok = df[df[status_col] == "ok"] if status_col in df else df
    g = ok.assign(_f=yes(ok[flag_col])).groupby("attack")["_f"]
    return (g.sum().astype(int).astype(str) + "/" + g.size().astype

aseal_ok = aseal.assign(status="ok")
craw_ok = craw.assign(status="ok")
syn = synthid.assign(det=synthid.detected.eq("yes"), part=synthid.d
syn_cell = syn.groupby("attack").apply(
    lambda g: f"{g.det.sum()}/{len(g)}" + (f" + {g.part.sum()} part
syn_cell["control_unmarked"] = syn_cell.pop("control_suno_unedited"

table4 = pd.DataFrame({
    "SynthID detected": syn_cell,
    "AudioSeal detected": count(aseal_ok, "detected"),
    "AudioSeal exact": count(aseal_ok, "payload_exact"),
    "WavMark exact": count(wavmark, "payload_exact"),
    "audiowmark exact": count(awm, "payload_exact"),
    "Perth detected": count(perth, "detected"),
    "AWARE detected": count(aware, "detected"),
    "AWARE exact": count(aware, "payload_exact"),
    "CRAW exact (30 s)": count(craw_ok, "majority_payload_exact"),
}).reindex(EDITS).fillna("not run")

def shade(v, row):
    if v == "not run" or "/" not in v:
        return "background-color:#EFEFEF"
    x, n = v.split(" ")[0].split("/")
    x, n = int(x), int(n)
    if row == "control_unmarked":
        return "background-color:#D9EAD3" if x == 0 else "backgroun
    if "partial" in v or 0 < x < n:
        return "background-color:#FFF2CC"
    return "background-color:#D9EAD3" if x == n else "background-co

table4.style.apply(lambda col: [shade(v, r) for r, v in zip(col.ind
```

Out[3]:

attack	SynthID detected	AudioSeal detected	AudioSeal exact	WavMark exact	audiowmark exact
none	not run	8/8	5/8	8/8	8/8
control_unmarked	0/1 (Suno)	0/8	0/8	0/8	0/8
mp3128	4/4	8/8	5/8	8/8	8/8
aac128	not run	8/8	4/8	not run	not run
gain6db	4/4	8/8	5/8	8/8	8/8
clip30	4/4	8/8	3/8	8/8	8/8
noise20db	3/4 + 1 partial	8/8	3/8	5/8	8/8
polarity	4/4	7/8	0/8	8/8	8/8
stemsplit	4/4	8/8	5/8	8/8	8/8
timestretch5	not run	0/8	0/8	not run	not run
speedup10	4/4	0/8	0/8	8/8	0/8
slowreverb	4/4	0/8	0/8	0/8	0/8
pitch2	4/4	0/8	0/8	0/8	0/8

Table 5. Wilson 95% intervals for eight songs

With only eight songs, a result like 5/8 is compatible with a true rate anywhere from about 31% to 86%. Only large differences (8/8 against 0/8) are clear.

In [4]:

```
def wilson(x, n, z=1.96):
    p = x / n
    d = 1 + z**2 / n
    centre = (p + z**2 / (2 * n)) / d
    half = z * math.sqrt(p * (1 - p) / n + z**2 / (4 * n**2)) / d
    return round(100 * max(0, centre - half)), round(100 * min(1, c

pd.DataFrame([{"result": f"{x}/8", "lower %": wilson(x, 8)[0], "upp
```

Out[4]:

	result	lower %	upper %
0	0/8	0	32
1	1/8	2	47
2	2/8	7	59
3	3/8	14	69
4	4/8	22	78
5	5/8	31	86
6	6/8	41	93
7	7/8	53	98
8	8/8	68	100

Table 6. ViSQOL v3 (audio mode, MOS-LQO) for AudioSeal and an MP3 encode

In [5]:

```
visqol = pd.read_csv(ROOT / "aseal/visqol_results.csv").pivot(index
visqol["MP3 loss"] = (visqol["identical"] - visqol["mp3128"]).round
visqol[["identical", "audioseal", "mp3128", "MP3 loss"]]
```

Out[5]:

comparison track	identical	audioseal	mp3128	MP3 loss
lyria_ambient_1	4.732	4.727	3.961	0.77
lyria_ambient_2	4.732	4.727	4.136	0.60
lyria_dense_1	4.732	4.728	4.488	0.24
lyria_dense_2	4.732	4.729	4.571	0.16
lyria_pop_1	4.732	4.728	4.382	0.35
lyria_pop_2	4.732	4.728	4.428	0.30
lyria_sparse_1	4.732	4.723	4.634	0.10
lyria_sparse_2	4.732	4.728	4.729	0.00

Appendix B. AWARE detection confidence by song and edit (✓ = all 20 bits recovered)

In [6]:

```
ok = aware[aware.status == "ok"].copy()
ok["cell"] = ok.detection_confidence.round(2).astype(str) + yes(ok.
appendix_b = ok.pivot(index="track", columns="attack", values="cell
appendix_b = appendix_b[[e for e in EDITS if e in appendix_b.column
appendix_b
```

Out[6]:

attack track	none	control_unmarked	mp3128	aac128	gain6db	clip30
lyria_ambient_1	1.0 ✓	0.02	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓
lyria_ambient_2	1.0 ✓	0.03	1.0 ✓	1.0 ✓	1.0 ✓	0.28 ✓
lyria_dense_1	1.0 ✓	0.03	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓
lyria_dense_2	1.0 ✓	0.04	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓
lyria_pop_1	1.0 ✓	0.02	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓
lyria_pop_2	1.0 ✓	0.02	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓
lyria_sparse_1	1.0 ✓	0.04	1.0 ✓	1.0 ✓	1.0 ✓	0.55 ✓
lyria_sparse_2	1.0 ✓	0.03	1.0 ✓	1.0 ✓	1.0 ✓	1.0 ✓

In [7]:

```
rows = []
for f in sorted(aware_run.glob("lyria_*/embedding.json")):
    e = json.loads(f.read_text())
    rows.append({"track": f.parent.name, "embed seconds": e["embed_
                 "SNR dB (16 kHz mono)": e["snr_db_16k_mono"], "cli
pd.DataFrame(rows)
```

Out[7]:

	track	embed seconds	SNR dB (16 kHz mono)	clipped samples
0	lyria_ambient_1	175.4	28.22	0
1	lyria_ambient_2	223.3	26.88	5
2	lyria_dense_1	237.8	20.30	4
3	lyria_dense_2	238.4	35.60	0
4	lyria_pop_1	211.1	13.62	6
5	lyria_pop_2	202.5	16.50	0
6	lyria_sparse_1	189.0	17.28	0
7	lyria_sparse_2	192.2	25.71	0

Why WAKE and IDEAW are not in Table 4

`check_model_availability.py` downloads each repository fresh and searches it for trained model files. Perth is included as a positive check (it should find a model). The newest result file is shown below.

In [8]:

```
avail = sorted((ROOT / "model_availability").glob("availability_*.c
print("File:", avail.name)
```

```
pd.read_csv(avail)[["checked_on", "watermark", "latest_commit_date"
```

```
File: availability_2026-11-10.csv
```

Out[8]:

	checked_on	watermark	latest_commit_date	mo
0	2026-11-10	WAKE	2025-05-28	
1	2026-11-10	IDEAW	2024-11-11	
2	2026-11-10	Perth	2026-09-01	src/perth/perth_net/pretrained/

AWARE run record

The settings file and the full Terminal output of the AWARE run, so the appendix shows exactly how the results were produced.

In [9]:
```
print((aware_run / "settings.json").read_text())
```

```
{
  "tracks": [
    "lyria_ambient_1.wav",
    "lyria_ambient_2.wav",
    "lyria_dense_1.wav",
    "lyria_dense_2.wav",
    "lyria_pop_1.wav",
    "lyria_pop_2.wav",
    "lyria_sparse_1.wav",
    "lyria_sparse_2.wav"
  ],
  "attacks": [
    "none",
    "control_unmarked",
    "mp3128",
    "speedup10",
    "slowreverb",
    "pitch2",
    "noise20db",
    "clip30",
    "gain6db",
    "stemsplit",
    "polarity",
    "timestretch5",
    "aac128"
  ],
  "payload_bits": 20,
  "threshold": 0.5,
  "aware_version": "2.0.1",
  "torch_version": "2.6.0",
  "seed": 42,
  "audio_method": "Embed in 16kHz mono; resample watermark differenc
e and add to both original stereo channels. Save PCM24. Apply shared
edits at original sample rate. Downmix and resample to 16kHz for det
ection.",
  "attacks_sha256": "21becd0e23a66b5a48b7454780f7d34a80a4d1876bf80ab
ab38624d0932148a9"
}
```


In [10]:
```
log = aware_run / "terminal_log.txt"
print(log.read_text() if log.exists() else "terminal_log.txt not fo
```

```
(base) Mac:watermark-stress-test-main antoniataubert$ conda activate
wm
(wm) Mac:watermark-stress-test-main antoniataubert$ python run_awar
e.py --track lyria_ambient_1.wav lyria_ambient_2.wav lyria_dense_1.w
av lyria_dense_2.wav lyria_pop_1.wav lyria_pop_2.wav lyria_sparse_1.
wav lyria_sparse_2.wav

Embedding: lyria_ambient_1.wav (AWARE optimizes the mark, this takes
a while)
  embedded in 175.4 s, SNR 28.2 dB
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.0166 detected=False exact_payload=False
```

```
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0427 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0406 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0372 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=0.9721 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.9988 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9998 detected=True exact_payload=True

Embedding: lyria_ambient_2.wav (AWARE optimizes the mark, this takes
a while)
  embedded in 223.3 s, SNR 26.9 dB
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.034 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0374 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0442 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.024 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=0.9875 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.2818 detected=False exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=0.9999 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9999 detected=True exact_payload=True

Embedding: lyria_dense_1.wav (AWARE optimizes the mark, this takes a
while)
  embedded in 237.8 s, SNR 20.3 dB
```

```
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.031 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0266 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0273 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0453 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9999 detected=True exact_payload=True

Embedding: lyria_dense_2.wav (AWARE optimizes the mark, this takes a
while)
  embedded in 238.4 s, SNR 35.6 dB
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.038 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0308 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0601 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0688 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=0.9622 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.9993 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9999 detected=True exact_payload=True
```

```
Embedding: lyria_pop_1.wav (AWARE optimizes the mark, this takes a w
hile)
  embedded in 211.1 s, SNR 13.6 dB
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.0174 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0203 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0231 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.021 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.9996 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9999 detected=True exact_payload=True

Embedding: lyria_pop_2.wav (AWARE optimizes the mark, this takes a w
hile)
  embedded in 202.5 s, SNR 16.5 dB
  Running none...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.0209 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0218 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0359 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0286 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.9999 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=1.0 detected=True exact_payload=True
```

```
  Running timestretch5...
    status=ok confidence=1.0 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9999 detected=True exact_payload=True

Embedding: lyria_sparse_1.wav (AWARE optimizes the mark, this takes
a while)
  embedded in 189.0 s, SNR 17.3 dB
  Running none...
    status=ok confidence=0.9998 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.0405 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=0.9995 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0318 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0838 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0316 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=0.7201 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.5484 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=0.9998 detected=True exact_payload=True
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=0.9998 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=0.9994 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9993 detected=True exact_payload=True

Embedding: lyria_sparse_2.wav (AWARE optimizes the mark, this takes
a while)
  embedded in 192.2 s, SNR 25.7 dB
  Running none...
    status=ok confidence=0.9999 detected=True exact_payload=True
  Running control_unmarked...
    status=ok confidence=0.0301 detected=False exact_payload=False
  Running mp3128...
    status=ok confidence=0.9998 detected=True exact_payload=True
  Running speedup10...
    status=ok confidence=0.0383 detected=False exact_payload=False
  Running slowreverb...
    status=ok confidence=0.0286 detected=False exact_payload=False
  Running pitch2...
    status=ok confidence=0.0557 detected=False exact_payload=False
  Running noise20db...
    status=ok confidence=0.7239 detected=True exact_payload=True
  Running clip30...
    status=ok confidence=0.9994 detected=True exact_payload=True
  Running gain6db...
    status=ok confidence=0.9999 detected=True exact_payload=True
```

```
  Running stemsplit...
    status=error confidence= detected= exact_payload=
  Running polarity...
    status=ok confidence=0.9999 detected=True exact_payload=True
  Running timestretch5...
    status=ok confidence=0.9997 detected=True exact_payload=True
  Running aac128...
    status=ok confidence=0.9994 detected=True exact_payload=True

Results saved to:
/Users/antoniataubert/Downloads/watermark-stress-test-main/aware/run
s/20261110_183744_325036/results.csv
(wm) Mac:watermark-stress-test-main antoniataubert$
```

Notes

- Perth results in `perth/claude_run_2026-11-10/` were produced by Claude with the group's unchanged `run_perth.py`.
- SynthID cannot be run in code. Its column reads the hand-logged Gemini checks in `synthid_results.csv`.
- Stem split is missing for AWARE and Perth until Demucs is installed in `.venv-demucs`.